FORTESCUE · MINE-TO-PORT GRADE CONTROL

Every train to spec

Know each train's lab grade when it leaves the load-out, and blend the next one on corrected stockpile grades.

73% → <5%

Solomon trains below typical Fe, blending on lab-reconciled grade (simulated)

11% → 71%

below-typical trains flagged at departure, 24-48 h before the lab result

US$2.8M/yr

grade penalties avoided, planning case (US$5.6M/yr simulated upper bound)

Synthetic data · illustrative penalty schedule

databricks

THE PROBLEM

Trains are blended on biased grades and checked 36 h later

Pit
Ore trucked to ROM

ROM stockpile
Block model grade **over-states Fe** by up to 0.45%

Train load-out
Controller picks the blend; cross-belt analyser **drifts**

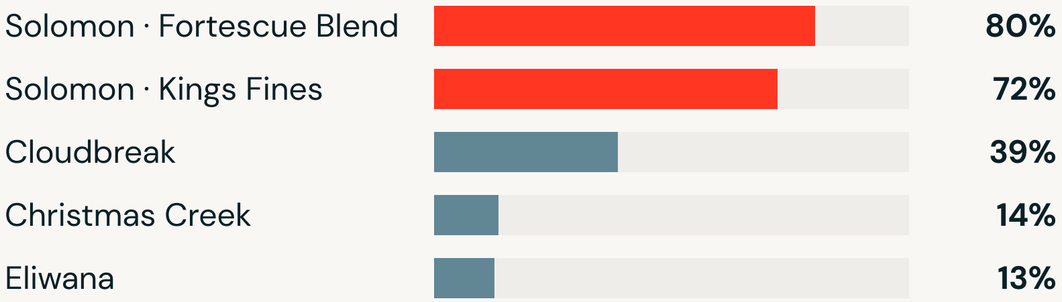
Rail
~34.5 kt per train, ~15 trains a day

Herb Elliott Port
Lab assay arrives; re-blend or rehandle

Vessel
Off-spec cargo pays the penalty

24-48 h between departure and the lab result

Trains below typical Fe, last 90 days

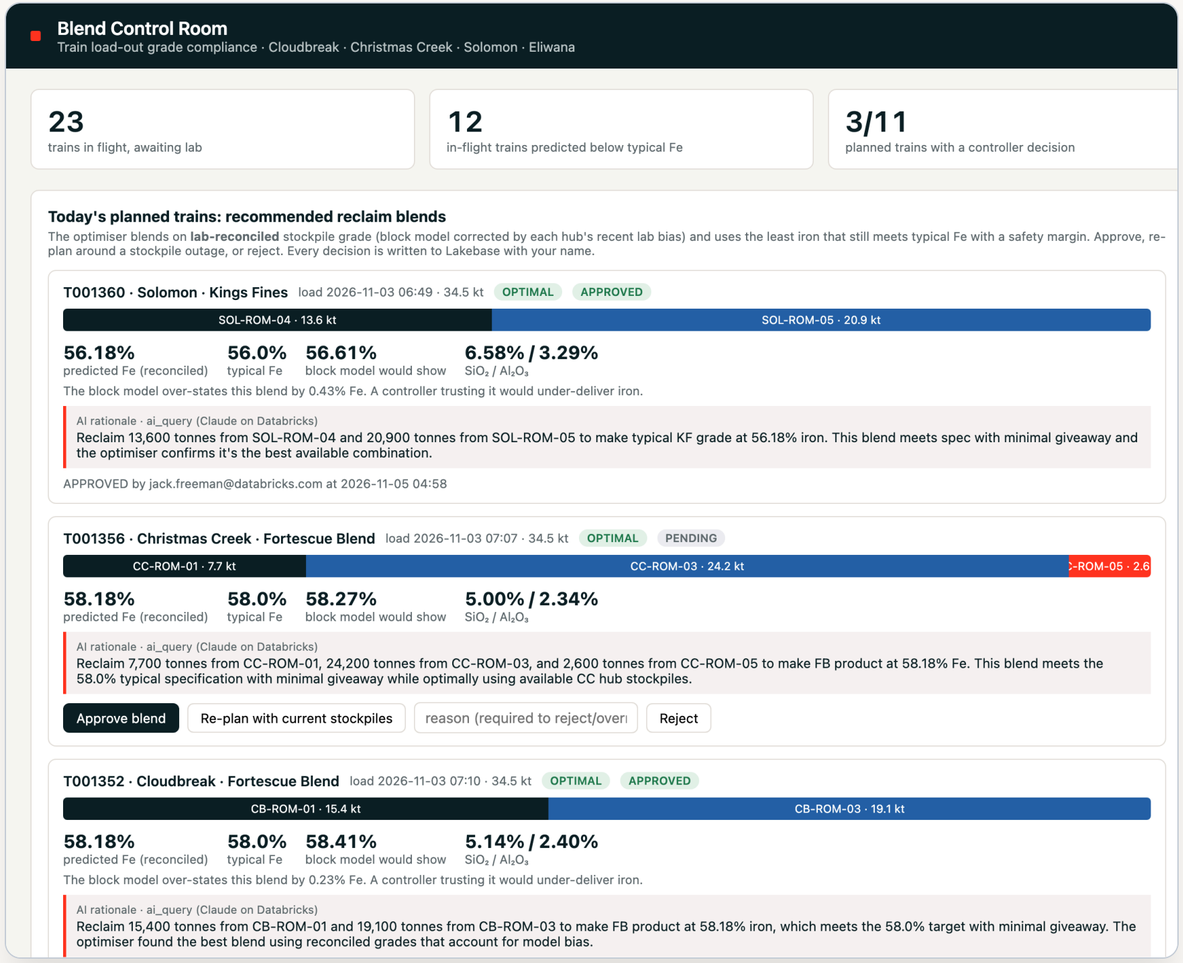


The bias is systematic, so it is correctable. **Solomon's block model reads 0.45% Fe high**; controllers trusting it under-deliver iron on most trains.

Hubs with little bias over-deliver instead: iron given away that could have been sold.

WHAT CHANGES

One control room for every train, on governed data



01 **Predicted lab grade for every train in flight**
ML model registered in Unity Catalog, scored each shift

02 **Recommended blend for every planned train**
Optimiser on lab-reconciled grades, with an AI-written reason

03 **Approve, re-plan around an outage, or reject**
Decisions written to Lakebase in milliseconds, with name and reason

04 **Ask questions in plain English**
Genie answers from certified metrics, under the user's own access

BUILT END TO END ON DATABRICKS

Lakeflow · Unity Catalog · MLflow · AI Functions · Lakebase · Genie · Databricks Apps

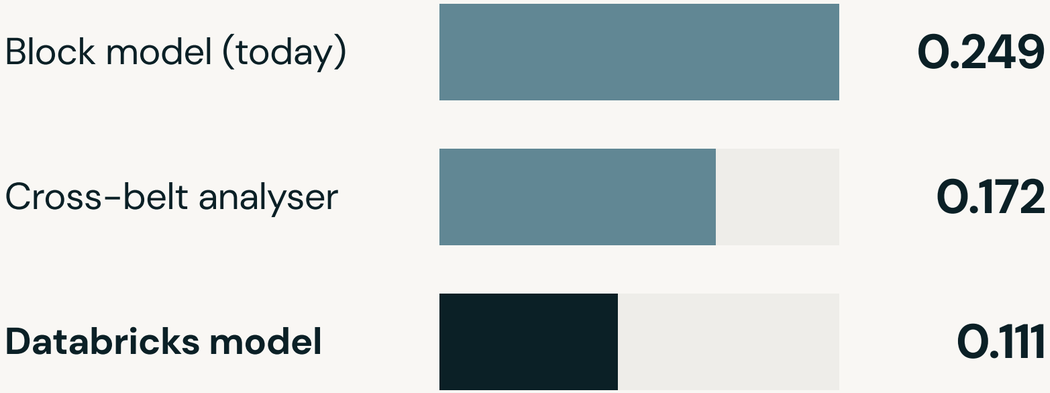
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DOES IT WORK

The grade is known at departure, not after the lab

Fe prediction error at departure

Mean absolute error vs lab assay, % Fe. Lower is better.



Below-typical trains caught at departure

Share of 131 below-typical trains flagged. Higher is better.



Held out: last 21 days, 313 trains the model never saw

Precision: 87%, fewer false alarms than the analyser

No look-ahead: only lab results returned before departure

databricks

WHAT IT IS WORTH

Revenue protected for planning, fewer surprises on shift

EXECUTIVE SPONSOR · GM INTEGRATED PLANNING & MARKETING

US$2.8M/yr

grade penalties avoided, planning case at 50% realisation

Vessels below typical Fe: 16 of 76 → 0 in the 30-day backtest

Trains below typical Fe: 39% → 6%

One certified definition of compliance for marketing, planning and the hubs

DOMAIN OWNER · SUPERINTENDENT GRADE CONTROL

36 h → 0

wait to know a train's grade, prediction at departure

Re-plan in seconds when a reclaimer goes down

Controllers stay in charge; every override and reason is logged

Each hub sees its own trains; penalty dollars only for commercial roles

Simulated on 90 days of synthetic data with an illustrative penalty schedule. Trade-off: average train Fe rises 57.82% → 58.07%, so more high-grade ore is used at Solomon.

NEXT STEP

An 8-week pilot at Solomon, decided on value

Weeks 1-2

Connect

LIMS, fleet management, analyser and rail feeds through Lakeflow Connect

Weeks 3-4

Calibrate

Retrain on Fortescue history; agree contract penalty terms and certified KPIs

Weeks 5-7

Shadow

Controllers see recommendations beside current practice; we measure the gap

Week 8

Decide

Value review against the baseline; go or no-go for all four hubs

SUCCESS LOOKS LIKE

- Below-typical trains at Solomon halved
- Prediction error under 0.15% Fe on Fortescue data
- No extra controller effort per shift

WE NEED FROM FORTESCUE

- Solomon grade control as pilot owner
- Read access to LIMS, FMS and analyser data
- Marketing input on penalty terms